Same page frame; a different orientation of the sky

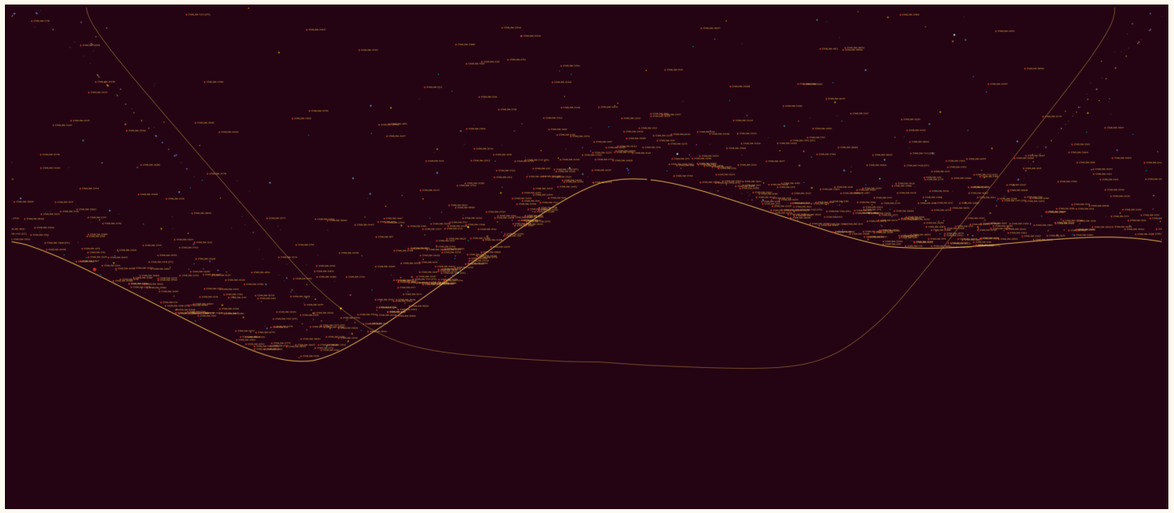
November 3 reference versus the current November 9 observer plate / San Francisco, 37.7749° N, 122.4194° W, 16 m

Executive Summary

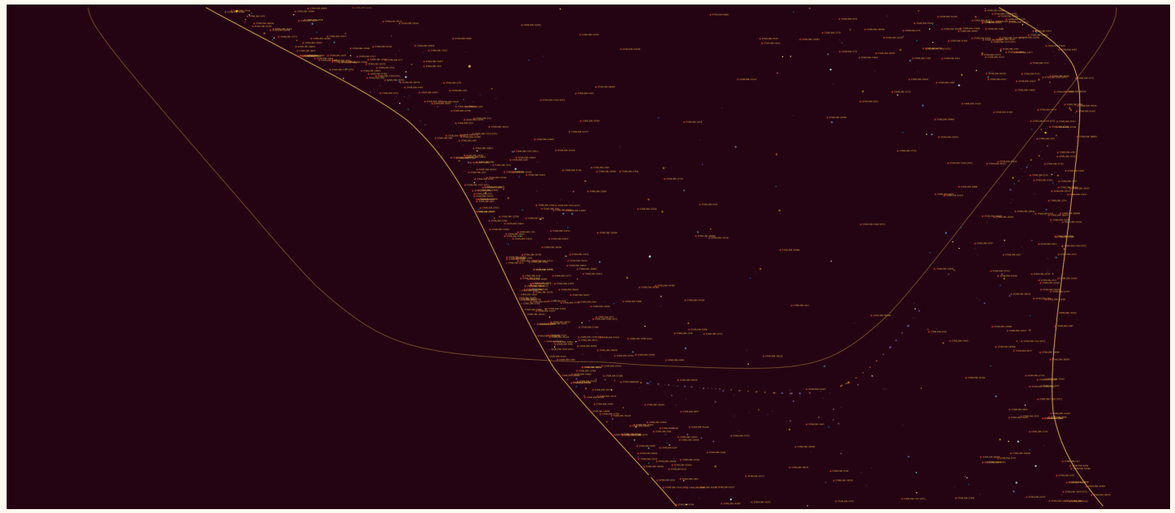
Both plates already use the same 4.35-inch printed cyclic move and the same 43.5-inch image width. The current framing is not rotated by 126°. The fixed celestial-equator paths match exactly. [1–3]

The observation instants are 5 days, 8 hours and 2 minutes apart. Earth’s rotation changes the observer’s orientation relative to the celestial grid by 125.758° modulo complete turns. Reprojecting the new local horizon produces a very different curve. [2–5]

NOV 3 / 17:10 UTC / 09:10 PST



NOV 9 / 01:12 UTC / NOV 8, 17:12 PST



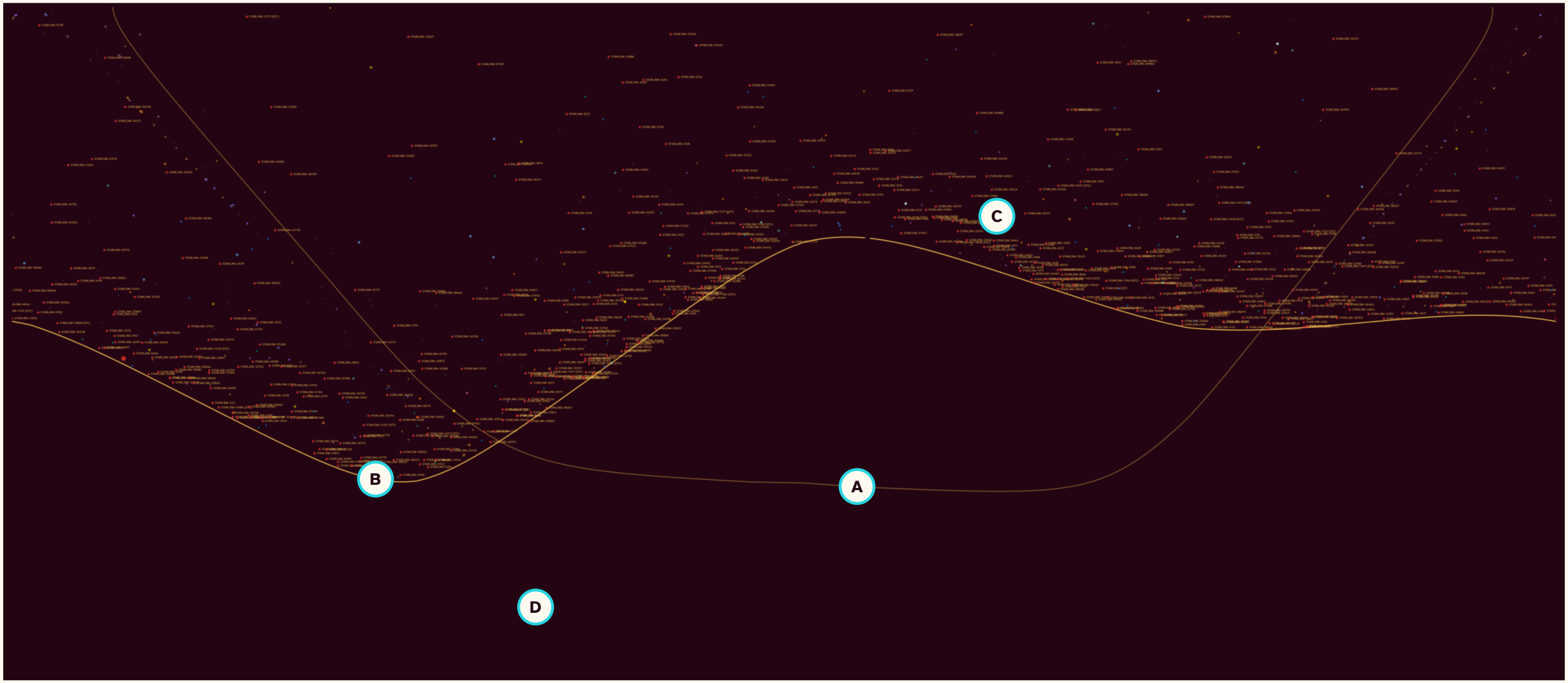
Why this is more than a calendar-date comparison

The local clock differs by 8 hours 2 minutes. At the same clock time exactly six days later, the residual sky rotation would be about 5.9°, after six complete turns. The six date numbers on the filenames are not six equal 24-hour steps at a fixed time of day.

The plotted population also changes: 1,088 to 972 objects above the horizon. My earlier explanation overstated the role of restored framing; the current two-plate comparison has identical framing.

Annotated November 3 reference

The supplied PNG is shown unchanged below; the date and counts come from the matching preserved editable SVG. [1,2]



A / Faint curve: celestial equator

Zero declination in the celestial coordinate grid. Its shape and placement are unchanged in the November 9 plate.

B / Bright curve: local horizon

The 0° elevation boundary for San Francisco at 17:10 UTC. This is a sky-selection boundary, not an orbital path.

C / Satellite marks and labels

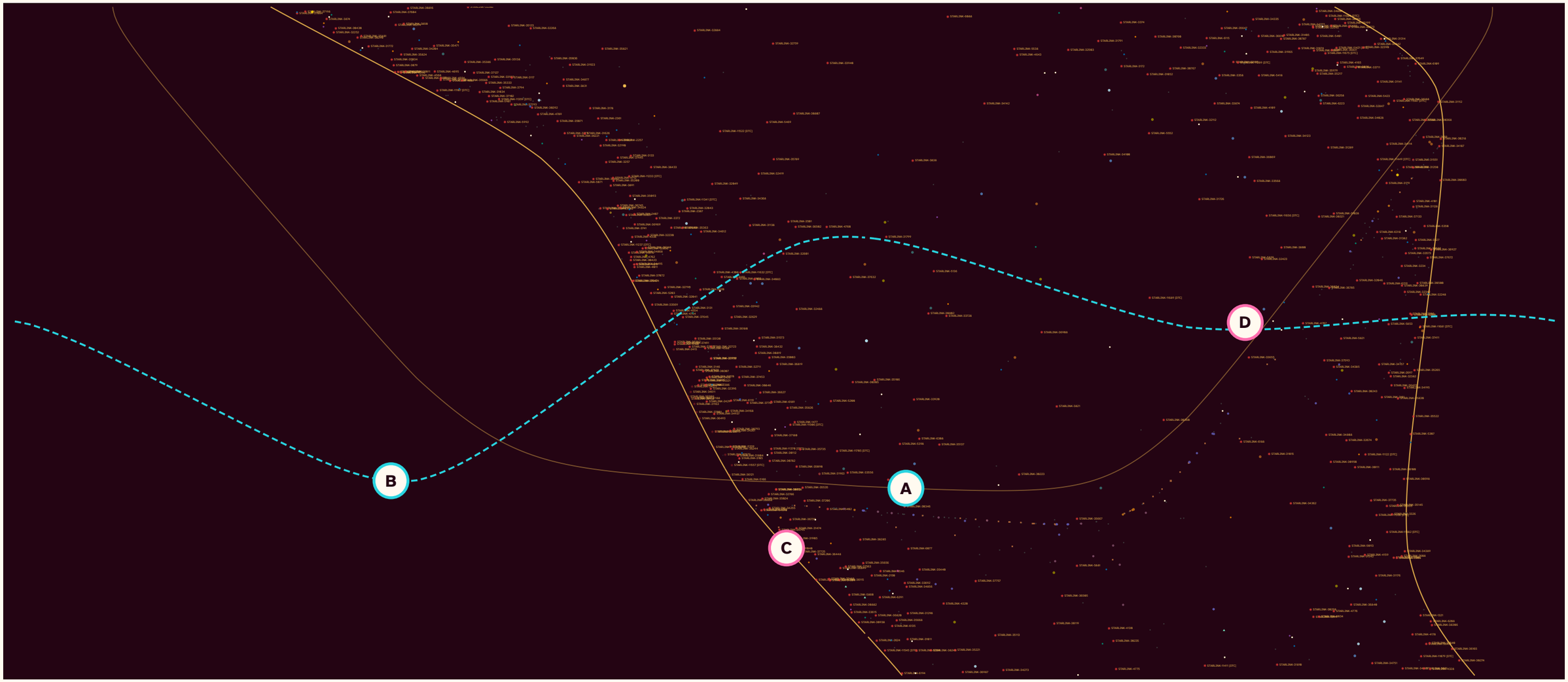
1,088 objects pass the above-horizon test, including 549 Starlink objects. The source is a calculated catalog view, not a photograph.

D / Empty part of the rectangle

Below-horizon directions remain in the full celestial frame but do not receive satellite marks. Empty space here is a selection effect.

November 9, annotated with the previous horizon

Current plate: gold horizon and current satellite positions. Cyan dashed overlay: November 3 horizon, placed in the identical frame. [2,3]



A / Equator stays registered

The reference equator is exactly the same SVG path geometry in both sources. This is the cleanest check that the sky grid did not move on the paper.

B / Previous horizon, cyan dashed

Only the old horizon is overlaid. Old satellite marks are omitted so a dashed line cannot be mistaken for a satellite trajectory.

C / Current horizon, gold

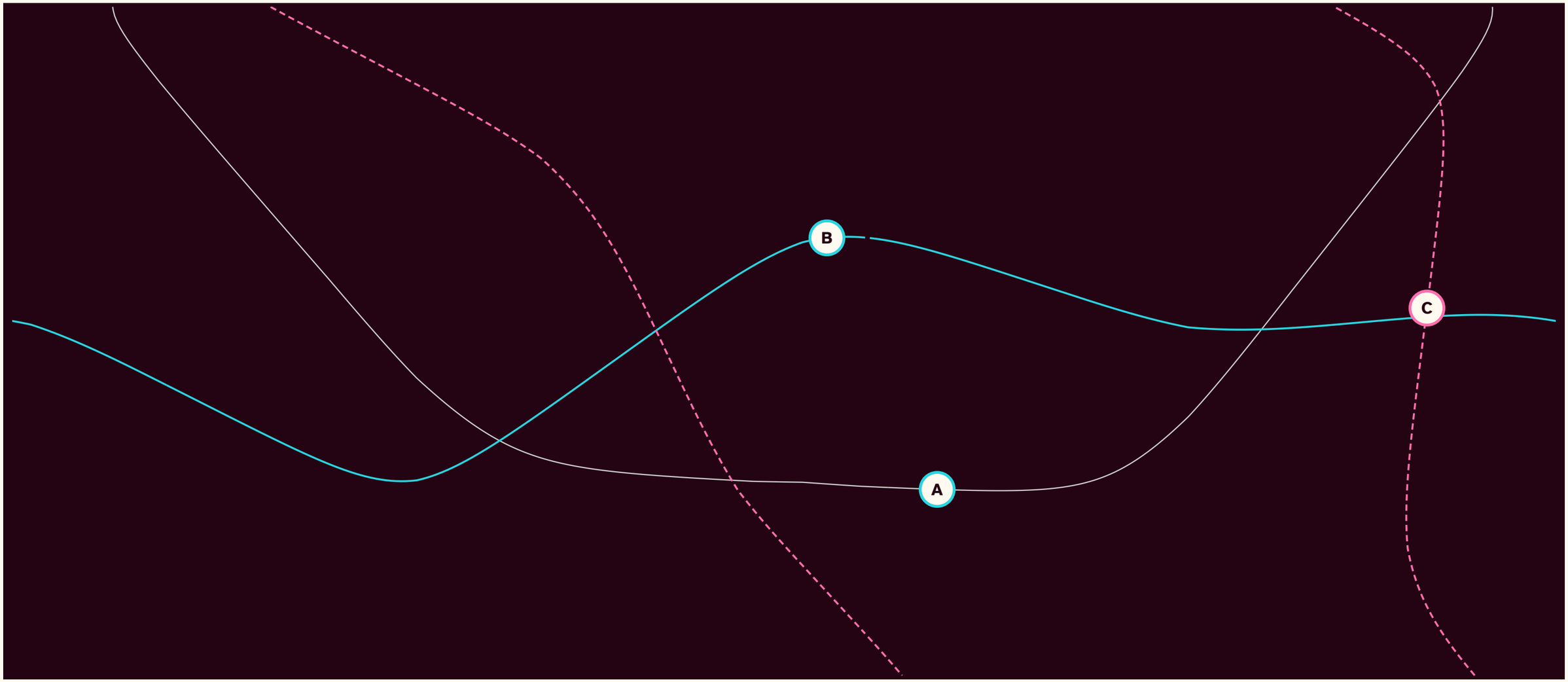
The later observation samples a different celestial hemisphere above the same San Francisco horizon. Curvature changes after AuthaGraph projection.

D / New satellite positions

972 objects are above the horizon, including 453 Starlink objects. Both their propagated positions and membership of the visible set can differ.

Remove the satellites: the horizon is what changes

Co-registered source curves: November 3 horizon in cyan; November 9 horizon in pink dashes; shared celestial equator in gray.



A / No frame adjustment was needed

Both source SVGs use scale 43.5 / 44 = 0.98863636, translation (0.25, 0.108253) inches, and three cyclic copies at −39.6, +4.4 and +48.4 native inches. Their clipping rectangle is 44 × 19.052559 native inches.

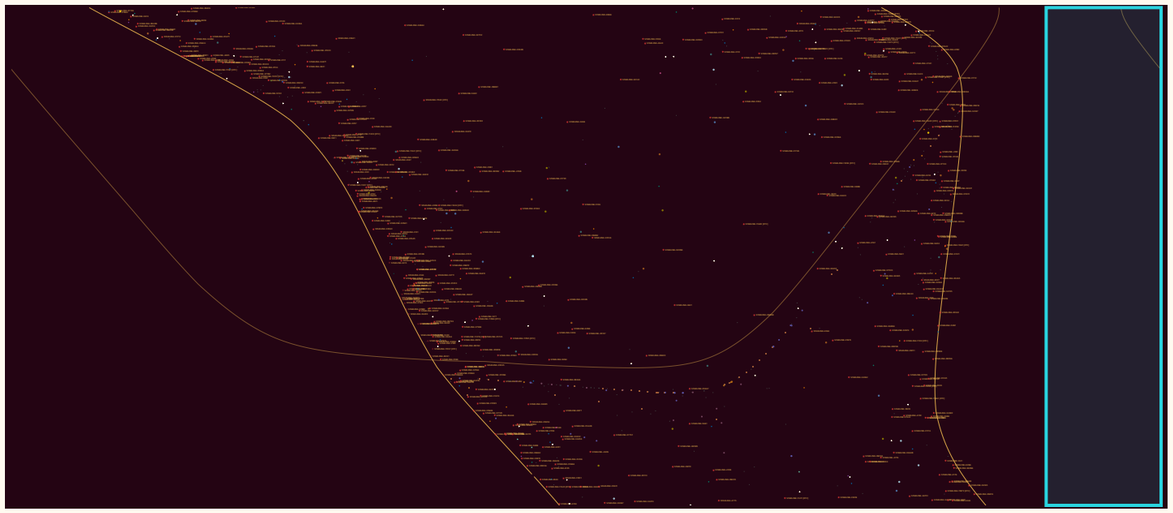
B–C / A moving boundary in a fixed celestial grid

The horizon is fixed relative to the observer's local ground, but its right ascension changes with Earth's rotation. AuthaGraph is nonlinear: rotating this great circle on the sphere can turn a broad wave into steep, split curves on the flat plate.

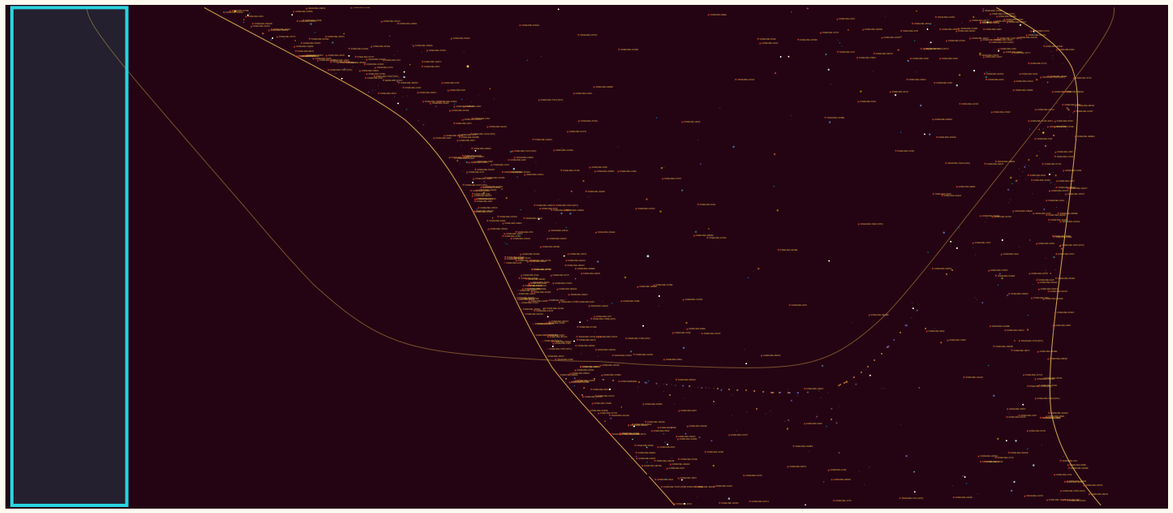
What cyclic framing alone actually does

Controlled comparison: both panels use the same November 9 source objects, horizon and time. Only the final horizontal display roll differs.

CONTROL / NO CYCLIC MOVE



CURRENT / 4.35-INCH PRINTED MOVE



The cyan strip is the rightmost 10% of the native rectangle; it reappears at the left edge after the roll.

Step 1 / Project celestial directions

Convert each satellite’s topocentric right ascension and declination into AuthaGraph coordinates. The central right ascension is 12 h in both source profiles. This stage produces a native rectangle 44 inches wide.

Step 2 / Move and wrap the already projected rectangle

Add 4.4 native inches to every x coordinate, wrap at 44, then print the rectangle at 43.5 / 44 scale. A point at native x = 2 moves to 6.4. A point at x = 42 moves to 2.4 after wrapping. Every y coordinate stays unchanged.

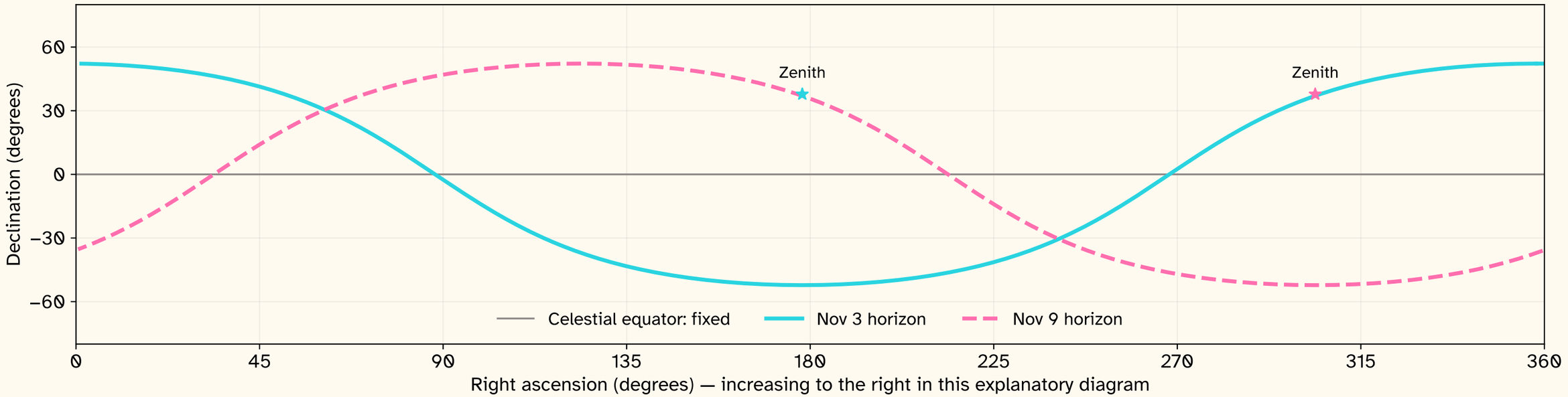
x_print = 0.25 + (43.5 / 44) × ((x_native + 4.4) mod 44)

y_print = 0.108253 + (43.5 / 44) × y_native

This is a 10% horizontal image roll, not 36° or 126° of physical sky rotation: AuthaGraph x is not a linear right-ascension axis. In the actual November 3 → November 9 comparison, this roll is already identical on both plates.

Why the same local horizon looks different

Earth rotates the observer relative to the celestial grid before projection; cyclic framing moves the finished rectangle after projection. [4,5]



Quantity	November 3	November 9
UTC calculation time	Nov 3, 17:10	Nov 9, 01:12
San Francisco clock (PST)	Nov 3, 09:10	Nov 8, 17:12
Model local sidereal orientation	178.055°	303.813°
Observer latitude / longitude	37.7749° N / 122.4194° W	same
Cyclic printed move	4.35 inches	same

The model sidereal phase changes by 125.758° after complete turns are removed. The horizon and zenith shift in right ascension; the equator remains at zero declination. “Above the horizon” means local elevation ≥ 0°, not necessarily above a curve on the printed page.

An exactly 24-hour sequence would leave about 0.986° of residual rotation, rather than return to an identical sky orientation. A sidereal day is about 23 h 56 min; satellite positions still would not repeat.

The populations change as well as their placement

Counts are unique plotted NORAD objects in each source, with the same ≥ 0° elevation threshold and the same observer location. [2,3]

Population / measure	Nov 3	Nov 9
Propagated global catalog	17909	17956
Above local horizon	1088	972
Starlink above horizon	549	453
Debris above horizon	53	47
Epoch exclusions	44	46
Sun altitude at observer	+24.617°	−2.351°

Only 202 NORAD IDs appear in both visible sets

886 appear only in the November 3 view; 770 appear only in the November 9 view. The net decrease of 116 visible objects is not evidence that 116 satellites disappeared. The global propagated catalog actually grows by 47 objects.

Time, orbital motion, the elevation filter and refreshed elements/catalog membership all contribute. These two endpoints alone do not isolate each contribution. Both plates include objects that are not naked-eye visible.

A useful reference object: AMAZONAS 3 / NORAD 39078

Coordinate	November 3	November 9
Local azimuth	108.390063°	108.386746°
Local elevation	13.759969°	13.763938°
Topocentric right ascension	245.877182°	11.634460°

This near-geostationary object stays at almost the same local azimuth and elevation. Its right ascension advances by about 125.757° modulo 360°, so its location on the fixed celestial-grid plate changes substantially. A large paper displacement does not necessarily mean a large change in local viewing direction.

Presentation change to keep separate: current debris x marks are white; the older plate used black x marks with halos. This changes visibility of small symbols, not the geometry of the horizon or equator.

Sources, limits and reproducibility

The overlays are explanatory additions. The published PNG and the current production SVG remain unchanged.

[1] Published reference image

https://benjamin.dekosnik.com/wp-content/uploads/2026/11/cartofreako-orbital-technosphere-observer-authagraph-44-19.052559-scaled.png

[2] Matching November 3 editable plate

output/msp-speculations-20261103/msp-speculation-5/wine-amber/orbital-technosphere-observer-authagraph-44-19.052559.svg.gz

[3] Current November 9 editable plate

output/starlink-star-light-world-20261110/maps/world/svg/slsl2-orbital-technosphere-observer.svg

[4] Renderer and Earth-rotation calculation

src.generate/orbiting-generation.h: horizontal_to_equatorial, add_reference_lines, visible_in_product; src.generate/orbiting-data.h: topocentric_equatorial; src.generate/third_party/sgp4/SGP4.cpp: gstime_SGP4.

[5] U.S. Naval Observatory / sidereal time

https://aa.usno.navy.mil/data/siderealtime — definition; https://aa.usno.navy.mil/faq/GAST — local-longitude relation and approximate GMST.

What was verified

Identical frame scales, translations and three cyclic-copy offsets; exact equality of all celestial-equator SVG paths; source-metadata counts reconciled to unique plotted NORAD IDs; matched-object coordinates read from both SVGs. The reference PNG visually matches the preserved November 3 artwork; its timestamp comes from the SVG metadata, not from the PNG filename.

Precision and interpretation

Sidereal values reproduce the renderer’s SGP4 GMST calculation using its UTC Julian-date input, plus San Francisco longitude. They describe this visualization; no higher-precision UT1 or apparent-sidereal correction is claimed. The RA/declination figure is an explanatory coordinate diagram, with zeniths marked by stars; it is not a new satellite propagation.

Inputs, full SHA-256 values, metadata, formulas and matched-object evidence are saved in evidence.json. Rebuild: python3 scripts/build-orbital-observer-comparison.py --reference PATH_TO_REFERENCE_PNG. The hourly sequence and 1080p MKV proposal are documented separately in docs/development/20261109_starlink_star_light_v4_observer_orientation_relative_celestial_grid.md.